INCH-POUND

A-A-54689
25 June 1992

COMMERCIAL ITEM DESCRIPTION

STERILIZER, SURGICAL DRESSING

(Pressure)

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for MIL-S-15786B which is cancelled.

This Commercial Item Description covers two types of pressure sterilizers, suitable for sterilizing surgical dressings, packs, and utensils aboard ship.

Type I - 20" x 36"

Type II - 16" x 24"

Salient characteristics:

The unit shall be an electrically heated, single door, stand mounted, cylindrical horizontal chamber, pressure sterilizer measuring 20" x 36" (Type I) or 16" x 24" (Type II). The pressure sterilizer shall be suitable for sterilizing surgical dressings, packs, and utensils aboard ship. Each sterilizer shall consist of, at least, a sterilizer body, door assembly, controls, generator, auxiliary components and accessories as prescribed herein.

The sterilizer body shall consist of a double shell, backheads, baffle plate, and end frame assembly. The backheads, baffle plate and end frame assembly shall be constructed of 304 stainless steel. The shell shall be of all welded construction, consist of two horizontal stainless steel chambers having a tensile strength of 50,000 to 65,000 psi, one telescoping the other to form a steam jacket over the inner shell. The sterilizer shall meet the requirements of ASME publication "Code for Unfired Pressure Vessels".

AMSC N/A FSC 6530

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:37Z
Check the source to verify that this is the current version before use.

The door assembly shall be dimensioned to assure a snug and proper fit into the opening of the sterilizing chamber. Design and construction shall permit changing easily the direction of the door swing without requiring additional parts and machine work. A locking device shall be provided which shall insure that when opening and closing the door the locking arms must fully engage before gasket seal can be established. A positive pressure operated lock shall be provided which shall be actuated by steam pressure within the chamber to permit unlocking the door only after all chamber pressure has been exhausted. The door shall also be provided with a means to permit tightening the door while locked; a door stop to permit the door to open only wide enough to allow unobstructed entrance to the sterilizing chamber; a rigid, adjustable door mounting designed for easy movement; and a suitably installed gasket. The door shall be manually operable, of suitable metal, and properly designed and machined to assure safe and effective operation.

A control device shall be provided which functions through a flexible bellows actuated by pressure within the sterilizing chamber, or by means of a thermostatic switch, either of which shall have a range of adjustment from 0-30 psig. Desired pressure shall be automatically maintained within plus or minus 2 psig. A single manually operated control handle shall actuate operating valve(s) in proper sequence to provide a complete sterilizing cycle including drying. As an alternate, individual manual valves shall be acceptable. The manual control shall permit the operator to admit steam into the sterilizing chamber for the sterilizing exposure, accomplish a slow exhaust, a fast exhaust, and a fast exhaust with drying. Accurate indicators to indicate the stage of each sterilizing cycle and completion of an entire cycle shall be provided. The sterilizer shall be equipped with a suitable dial type thermometer to accurately indicate the internal temperature of the chamber.

Auxiliary components shall include an exhaust system, chamber drain, valves, piping, heating system, and miscellaneous items such as chamber and jacket gauge, and necessary valves and fittings. The exhaust system shall include an atmospheric vent. A vacuum breaker shall be provided in the water supply line as a sanitation protective device. An exhaust connection shall be provided for attachment to the atmospheric vent piping. The atmospheric vent shall drain through an air break to atmosphere. The chamber drain line shall be large enough for the purpose intended, shall enter

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:37Z
Check the source to verify that this is the current version before use.

at the bottom front of the sterilizing chamber, and shall have an easily removable corrosion-resisting metal screen fitted over or into the drain outlet. A thermostatic trap shall be provided which allows passage of condensate from the sterilizing chamber, and expulsion of air. The sterilizer shall maintain a slight discharge and constant circulation of steam during a sterilizing cycle, and shall be suitable for operating at a pressure of at least 40 psi. Trap shall partially close against steam pressure, and shall be compensated for pressure and temperature. The thermostatic trap shall have a removable element, seat, and gasket, and a check valve with removable disc. Valves furnished with each sterilizer shall be of suitable metal with replaceable corrosion-resisting metal seats and Teflon discs or equivalent high quality functional parts to minimize maintenance. The sterilizer shall be leakproof when tested at a hydrostatic pressure of 300 psig on the shell and 150 psig on the seat. Valve handles shall be of heat-resistant, non-loosening plastic material, easily removable and replaceable, with the function of the valve clearly indicated on the handle in a legible, durable manner. Valve handles shall be conveniently located and operable from the front of the sterilizer. Valves shall open by turning counter-clockwise. Wall sleeves and collars shall be provided as required. The safety valve shall meet the requirements of the ASME and be set at the approved design pressure of the vessel. Piping provided with the sterilizer shall be brass or copper. Exposed pipes shall be chromium or nickel plated with a polished finish. Piping and fittings shall be designed to prevent back flow of polluted water or waste into the water supply system or the sterilizer. The sterilizer shall be equipped for heating by electricity and shall include an electric steam generator. The generator snall meet the requirements of ASME, and, shall be equipped with automatic fill. Shall consist of a generator tank, electric immersion heaters (enclosed type) low water cut-out with manual reset, and necessary switches, relays and connections contained in a suitable switch box, and the necessary valves and piping to connect to the water supply, the sterilizer jacket and chamber, and the waste line. The generator shall be the horizontal type and suitably constructed of corrosion-resisting metal. Electric heater(s) shall operate effectively on 440 volt, 3 phase, 60 cycle, AC, and shall be controlled by an indicating switch which shows when the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:37Z
Check the source to verify that this is the current version before use.

heater(s) are on. Under normal circumstances when in the "ON" position, the heater(s) shall operate continuously until reaching the desired pressure within a range of 15 to 20 psig, and shall cycle to maintain the preselected pressure, controlled by a pressure switch as previously described. Heater(s) and switch box shall be easily replaced and serviced. Shall have a maximum rating of 15KW. The low water cut-out shall be accurate, positive acting, and preclude operation of heater(s) if the water in the generator drops below a safe level.

Generator shall be equipped with an easily visible water gage to accurately show the water level within the generator. Electrical switches, contactors, and indicating devices shall be provided in a suitable switch box securely attached to the sterilizer. At least six (6) feet of electric power cord without plug shall be provided. The sterilizer shall meet the requirements of Underwriters' Laboratories Inc., as appropriate. An easily removable perforated alloy tray, properly sized to fit into the chamber and rest slightly above the chamber floor shall be provided.

The sterilizer shall be mounted within a rigid durable framework of corrosion-resisting steel. The sterilizer stand shall rest on floor plates.

The following information shall be permanently affixed, and located on the sterilizer to permit reading without difficulty.

a. Name of manufacturer.
b. Manufacturer's type and model number.
c. Serial number.
d. Voltage, amperage, and kilowatt rating.

The unit shall be capable of exhausting sterilizing chamber pressure from 20 psi to 0 in three minutes or less.

The sterilizer shall be capable of reaching a sterilizing temeprature of 250°F from a cold start in not more than 30 minutes.

When properly connected to the appropriate utility service lines and operated in accordance with the manufacturer's instructions, the sterilizer shall demonstrate the ability to accomplish a complete kill when tested as follows:

Source: https://assist.dla.mil - Downloaded: 2026-10-26T09:37Z
Check the source to verify that this is the current version before use.

a. Inoculate six paper strips having approximate dimensions of 2 inches by 1/2 with a spore population of Bacillus Stearothermophilus that will survive 250°F for five minutes, but will be killed when exposed to a temperature of 250°F for twelve minutes.

b. Place four of the inoculated strips into the sterilizing chamber by inserting each strip between 4 by 4 inch gauze sponges and locating two of the strips in the bottom front of the chamber and two in the bottom rear. Two strips shall be withheld for viability controls.

c. Operate the sterilizer in accordance with the manufacturer's instructions so that the test strips in the chamber are exposed to a temperature of from 250°F to 254°F for a period of twelve minutes, and then exhaust the sterilizing chamber by means of a fast exhaust.

d. Remove test strips from the sterilizing chamber in a manner that will preclude contamination, and test for sterility as recommended in the latest edition of the Pharmacopoeia of the United States.

The sterilizer shall be supplied with two copies of service data containing complete instructions for installation, operation, maintenance, a list of component parts, and wiring diagrams.

Waiver of testing. The tests specified herein may, at the procurement contracting officer's discretion, be waived for contracts which specify sterilizers conforming to the same size and general characteristics as those previously tested and for which the manufacturer has certified or recertified test records on file.

Workmanship. The sterilizer shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One sterilizer , as specified, constitutes one unit.

Source: https://assist.dla.mil - Downloaded: 2026-10-26T09:37Z
Check the source to verify that this is the current version before use.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 2.5 (percent defective) and an inspection level of II. The unit of product for examination purposes shall be one sterilizer as specified in this description.

Sampling for tests. Sampling for tests, including dimensional test, shall be conducted in accordance with MIL-STD-105, with an AQL of 2.5 (percent defective) and an inspection level of II. The unit of product for test purposes shall be one sterilizer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:37Z
Check the source to verify that this is the current version before use.

Tests.

1. Chamber evacuation - Shall evacuate chamber from 20 psi to 0 in three minutes.

2. Sterilization effectiveness - Shall accomplish a complete kill when tested as specified herein.

3. Pressure control accuracy - Shall maintain control set point plus or minus 2 psig.

4. Temperature come-up time - Starting with generator filled to within 1/2 inch of the top with water at 70°F, shall reach a sterilizing temperature of 250°F in not more than thirty minutes.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Source: https://assist.dla.mil - Downloaded: 2026-10-26T09:37Z
Check the source to verify that this is the current version before use.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document.

6530-00-065-7443 - Type I
6530-00-065-7444 - Type II

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:37Z
Check the source to verify that this is the current version before use.

MILITARY INTERESTS:

Custodians:

Air Force - 03
Army - MD
Navy - MS

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

FDA - MPQAS
USPHS
VA - OSS

Project Number: 6530-2129

Location: STERL1BC.DOC

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:37Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER: A-A-54689	2. DOCUMENT DATE (YYMMDD): 25 June 1992

3. DOCUMENT TITLE: STERILIZER, SURGICAL DRESSING (Pressure)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME DEFENSE PERSONNEL SUPPORT CENTER ATTN: DPSC-MSE	b. TELEPHONE (Include Area Code) (1) Commercial: AC 215-737-2870 (2) AUTOVON: AV 444-2870
c. ADDRESS (Include Zip Code) 2800 SO. 20TH STREET PHILADELPHIA, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 — Previous editions are obsolete — 9879v

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:37Z
Check the source to verify that this is the current version before use.